

# L-functions for all meromorphic $\mathrm{SL}_2(\mathbb{Z})$ modular forms, via manifestly finite contours

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## Abstract

We give a single manifestly finite, regulator-free functional realizing in one object L-functions quasi-periods and period polynomials of meromorphic modular forms on  $\mathrm{SL}_2(\mathbb{Z})$ , with generic poles throughout the upper half-plane  $\mathbb{H}$  and at the cusp of  $\mathrm{SL}_2(\mathbb{Z})$ . Additionally, our construction preserves holomorphicity. It is a two-segment contour integral,  $L^*(f, s) = L_T^*(f, s; \gamma_{\tau_0}^T) + L_S^*(f, s; \gamma_{\tau_0}^S)$ , where  $L_T^*$  and  $L_S^*$  have distinct integration kernels, and  $\gamma_{\tau_0}^S$  is a path in  $\mathbb{H}$  from  $S(\tau_0)$  to  $\tau_0$ , and the same for  $\gamma_{\tau_0}^T$ . For weakly holomorphic forms our L-function is independent of  $\tau_0$  and of the path, and reproduces classical L-functions for holomorphic modular forms, reproduces the L-functions of weak modular forms in [1], which previously needed non-holomorphic regularization to be defined, and reproduces Brown's quasi-periods [2]. For properly meromorphic modular forms with poles inside the interior of  $\mathbb{H}$ , our L-function depends on the homotopy classes, defined in terms of pole-crossings. There is a natural choice of trivial homotopy class, which renders our L-functions canonical when the modular form has a pole in the interior of the fundamental domain, generalizing [3].

## 1 Introduction and preliminary concepts

Our primary goal in this note is to put forth a unifying generalization of the L-function associated to a given modular form. Our L-function naturally encompasses the classical case of L-functions associated to holomorphic modular forms [4], and seamlessly unifies Ref. [1]'s extension to L-functions of modular forms with poles at the cusp with Ref. [3]'s extension to L-functions of modular forms with poles at finite locations, exclusively away from cusps. Secondly, we show that our L-functions reproduce the numerical results for quasi-periods associated to weak cusp forms [2].

*Note:* Throughout this document, we refer to L-integrals and L-functions. The classical L-function-as-Dirichlet-series which has been suitably extended to the complex  $s$  plane is simply not defined when the modular form that defines the L-function has poles either at the cusp or at finite locations within the upper half-plane. So, if there is ever a doubt, we state clearly: *any L-function in this note is defined by an L-integral*. The focus of this note is to define L-integrals for any integer weight  $\mathrm{SL}_2(\mathbb{Z})$  modular form which may have poles at cusps or away from cusps.

### 1.1 Modular forms

Write  $\mathrm{Re}(z)$  and  $\mathrm{Im}(z)$  for the real and imaginary parts of a complex number  $z \in \mathbb{C}$ . Let  $\mathbb{H} := \{\tau \in \mathbb{C} \mid \mathrm{Im}(\tau) > 0\}$  denote the upper half-plane.  $\mathrm{SL}_2(\mathbb{Z})$  is the group generated by  $T := \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  and  $S := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ .

There is a natural  $\mathrm{SL}_2(\mathbb{Z})$  action on functions  $f : \mathbb{H} \rightarrow \mathbb{C}$ , graded by integers  $k$ :  $(f|_k \gamma)(\tau) := (c\tau + d)^{-k} f(\gamma \cdot \tau)$  for all  $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ . Similarly,  $\gamma \cdot \tau := \frac{a\tau + b}{c\tau + d}$ .

In this vein, a *holomorphic modular form* of weight  $k$  for  $\mathrm{SL}_2(\mathbb{Z})$  is a holomorphic function  $f : \mathbb{H} \rightarrow \mathbb{C}$  such that  $(f|_k \gamma)(\tau) = f(\tau)$  for all  $\gamma \in \mathrm{SL}_2(\mathbb{Z})$  (aka  $f(\tau + 1) = f(-\frac{1}{\tau})\tau^{-k} = f(\tau)$ ) for all

$\tau \in \mathbb{H}$ , and  $f(\tau) = \mathcal{O}(1)$  as  $\text{Im}(\tau) \rightarrow \infty$ , and which has no poles at finite locations within  $\mathbb{H}$ . By virtue of the periodicity  $f(\tau) = f(\tau + 1)$  and holomorphicity, each holomorphic modular form has a  $q$ -series decomposition,

$$f(\tau) = \sum_n c_f(n) e^{2\pi i n \tau} = \sum_n c_f(n) q^n \quad , \quad q := e^{2\pi i \tau} \quad . \quad (1)$$

In order for  $f$  to be bounded as  $\tau \rightarrow i\infty$ , we conclude that  $c_f(n) = 0$  for all  $n < 0$ , i.e. that  $f = \sum_{n \geq 0} c_f(n) q^n$ ; for such modular forms  $f$  we write  $f \in M_k$ . If a modular form  $f$  additionally has a vanishing constant term, i.e. if  $c_f(0) = 0$ , then  $f$  is a *cusp form*, and we write  $f \in S_k$ . Clearly,  $S_k \subset M_k$ . Further,  $M_k$  is a finite-dimensional vector-space, as is  $S_k$ . Holomorphic  $\text{SL}_2(\mathbb{Z})$  modular forms must have even integer weights, and  $k \geq 4$  [4].

A product of modular forms of weights  $k_1$  and  $k_2$  is a modular form of weight  $k_1 + k_2$ , so  $M_* := \bigoplus_k M_k$  is naturally a (graded) ring. This graded ring is freely generated by the two modular forms  $E_4 \in M_4$  and  $E_6 \in M_6$ , the  $k = 4, 6$  specializations of the *weight  $k$  Eisenstein series*  $E_k$ , which may be defined as follows:

$$E_k := \frac{1}{2} \sum_{(m,n)=1} \frac{1}{(m+n\tau)^k} = 1 - \frac{2k}{B_k} \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n \quad , \quad (2)$$

where  $\sum_{(m,n)=1}$  denotes summing over all relatively prime  $(m, n) \in \mathbb{Z}^2$ , the Bernoulli number  $B_n$  is the coefficient of  $t^n/n!$  of the Laurent expansion of  $t/(e^t - 1)$  about  $t = 0$ , and  $\sigma_m(n) := \sum_{d|n} d^m$  is the Divisor function where  $d|n$  means  $n/d \in \mathbb{N}$ . In short,  $M_* = \mathbb{C}[E_4, E_6]$ . We now define the *modular discriminant*:

$$\Delta := \frac{E_4^3 - E_6^2}{1728} = q + \mathcal{O}(q^2) \in S_{12} \quad . \quad (3)$$

Holomorphicity is most canonically relaxed via allowing a modular form to have a pole at the cusp of  $\text{SL}_2(\mathbb{Z})$ , i.e. for a weight  $k$  modular form for  $\text{SL}_2(\mathbb{Z})$  to have negative powers of  $q$  in its  $q$ -series expansion. In this case,  $f$  diverges as  $\tau \rightarrow i\infty$  as  $f \sim e^{-2\pi N \tau}$ , where  $-N$  is the largest negative number such that  $c_f(n)$  is non-zero. Such a modular form  $f$  is called a *weak modular form*, and we write  $f \in M_k^!$ . If a weak modular form  $f$  has a vanishing coefficient of  $q^0$ , i.e. if  $c_f(0) = 0$ , it is called a *weak cusp form* and we write  $f \in S_k^!$ . Clearly,  $S_k^! \subset M_k^!$ ,  $S_k \subset S_k^!$  and  $M_k \subset M_k^!$ . Two important weak cusp forms in this note are as follows:

$$J := \frac{E_4^3}{\Delta} - 744 = \frac{1}{q} + \mathcal{O}(q) \in S_0^! \quad , \quad (4)$$

$$\widehat{\Delta} = (J^2 + 24J - 393444)\Delta := \frac{1}{q} + \mathcal{O}(q^2) \in S_{12}^! \quad . \quad (5)$$

In this work, we further relax from even weakly holomorphic modular forms, and consider *meromorphic modular forms* of weight  $k$  which are quotients:  $f = \frac{f_1}{f_2}$  where  $f_i \in M_{k_i}$ , the denominator  $f_2$  is not identically zero, and  $k = k_1 - k_2$ . We write  $F_k$  for the vector space of meromorphic modular forms of weight  $k$ , and set  $F_* := \bigoplus_k F_k$ . Then  $F_k$  vanishes unless  $k$  is an even integer, and  $F_*$  may be regarded as a “graded field of fractions” of  $M_*$ .

## 1.2 L-functions for holomorphic modular forms and periods for cusp forms

For any  $f \in S_k$ , one may define the associated L-function:

$$L(f, s) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s) \stackrel{f \in S_k}{:=} \frac{(2\pi)^s}{\Gamma(s)} \int_0^\infty f(it) t^{s-1} dt = \sum_{n \geq 1} \frac{c_f(n)}{n^s} \quad . \quad (6)$$

A similar definition applies for  $f \in M_k$ . (The classical definition of  $L^*(f, s)$  for  $f \in M_k$  is the focus of Remark 3.6, below.) Similarly, one may define a period polynomial for  $f \in S_k$  in terms of this L-function,

$$r_f(X, Y) := (2\pi i)^{k-1} \sum_{\ell=0}^{k-2} (-1)^\ell \binom{k-2}{\ell} i^{\ell+1} L^*(f, \ell+1) X^{k-2-\ell} Y^\ell. \quad (7)$$

This polynomial,  $r_f(X, Y) \in V_{k-2} := \mathbb{C}[X, Y]_{k-2}$ , has a natural action under  $\mathrm{SL}_2(\mathbb{Z})$ ,  $(P|_\gamma)(X, Y) = P(aX + bY, cX + dY)$  for  $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ . This period polynomial can be decomposed into a basis of highly constrained polynomials, which are annihilated by  $|(1+T)$  and  $|(1+U+U^2)[6]$ , where  $U := TS$  generates an order-3 cyclic subgroup of  $\mathrm{SL}_2(\mathbb{Z})$ . We define the *periods* of a cusp form to be coefficients of  $r_f$  within this highly constrained sub-space  $W$  of  $V_{k-2}$ , i.e.  $W \subset V_{k-2}$ .

**Why this work?** As classically computed, these L-functions and associated period polynomials (when defined in terms of the special values of the associated L-function) are not defined when  $f$  has poles either at the cusp of  $\mathrm{SL}_2(\mathbb{Z})$  or within the interior of  $\mathbb{H}$ . As stated above, there are extant, but partial and non-overlapping, extensions of the definition of  $L(f, s)$  to  $f \in M_k^!$  [1] or  $f \in F_k$  when  $f$  is regular at the cusp of  $\mathrm{SL}_2(\mathbb{Z})$  [3]. In this note, we provide a construction that unifies these disparate extensions. We start with generic L-integrals in section 2, then specialize to  $f \in M_k^!$  in section 3, and finally explore the more general context of  $f \in F_k$  in section 4.

## 2 A finite L-integral

**Definition 2.1.** Let  $\tau_0 \in \mathbb{H}$  and  $f \in F_k$ , let  $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ ,  $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  denote the generators of  $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ , and let  $f^{-1}(\infty)$  be the locus of all poles of  $f \in \mathbb{H}$  (inclusive of the cusp of  $\mathrm{SL}_2(\mathbb{Z})$ ). The  $S$ - and  $T$ -segments at  $\tau_0$  are smooth paths in  $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$  from  $\sigma^{-1}\tau_0$  to  $\tau_0$ ,

$$\gamma_{\tau_0}^S : -1/\tau_0 \longrightarrow \tau_0, \quad \gamma_{\tau_0}^T : \tau_0 - 1 \longrightarrow \tau_0. \quad (8)$$

Avoiding poles is automatic for  $f \in M_k^!$ , as long as the base-point  $\tau_0$ , is not at a cusp of  $\mathrm{SL}_2(\mathbb{Z})$ , i.e. there is only the trivial homotopy class. When  $f \in F_k$ , and  $f$  has poles at finite and non-zero imaginary heights, integration contours fall into nontrivial homotopy classes.

**Definition 2.2.** Let  $\zeta(s, x) = \sum_{m \geq 0} (m+x)^{-s}$  as a function  $\zeta : \mathbb{C}_s \times (\mathbb{C} \setminus \mathbb{N})_x \rightarrow \mathbb{C}$  denote the Hurwitz zeta function (meromorphic in  $s$ , holomorphic in  $x \in \mathbb{H}$ ). We define Hurwitz  $T$ -kernel as

$$\tilde{k}_T(\tau, s) := \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(1-(k-s), \tau+1). \quad (9)$$

**Definition 2.3** (Generic L-integral). Let  $f \in F_k$ ,  $\tau_0 \in \mathbb{H}$ , and  $s \in \mathbb{C}$ . Further, let  $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$  be the  $S$ - and  $T$ -segments at  $\tau_0$ , as in Definition 2.1, and let  $k_T$  be as defined in Definition 2.2. The L-integral of  $f$  is

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := L_S^*(f, s; \gamma_{\tau_0}^S) + L_T^*(f, s; \gamma_{\tau_0}^T) := e^{-\pi i s/2} \left( \int_{\gamma_{\tau_0}^S} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, s) d\tau \right) \quad (10)$$

We use this  $L$ -integral to define a generalized  $L$ -function:

**Definition 2.4** (Generic L-function). Let  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  be as defined in Definition 2.3. We define the generalized L-function of  $f \in F_k$  to be,

$$L(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T). \quad (11)$$

Having defined the L-integral and L-function, we prove they are manifestly finite for  $f \in F_k$ :

**Lemma 2.5** (Finiteness of L-integrals). *For  $f \in F_k$ , both  $L_S^*(f, s; \gamma_{\tau_0}^S)$  and  $L_T^*(f, s; \gamma_{\tau_0}^T)$  in Definition 2.3 are each, separately finite for every  $s \in \mathbb{C}$ , every  $\tau_0 \in \mathbb{H}$ , and for every path  $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$ .*

*Proof.* Both segments are compact and lie in  $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ , so  $f$  is continuous on them;  $\tau^{s-1}$  is continuous on  $\mathbb{H}$ , and  $\tilde{k}_T(\tau, s)$  is continuous there since  $\zeta(\nu, \tau + 1)$  is holomorphic in  $\tau \in \mathbb{H}$  (Definition 2.2). A continuous integrand on a compact path is integrable, for every  $s \in \mathbb{C}$ .  $\square$

We now turn to path-dependence and  $\tau_0$ -independence of the L-integral and the L-function of  $f \in F_k$ . This holds for all  $f \in M_k^!$ :

**Proposition 2.6** (Basepoint & path independence for  $f \in M_k^!$ ). *For  $f \in M_k^!$  the value  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  is independent of  $\tau_0 \in \mathbb{H}$  and of the paths: with no interior poles,  $\mathbb{H}$  is simply connected, so all admissible contours are homotopic. We write  $L^*(f, s)$  for the common value and  $L(f, s) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s)$ .*

*Proof.* With no interior poles the integrands  $f(\tau)\tau^{s-1}$  and  $f(\tau)\tilde{k}_T(\tau, s)$  are holomorphic on  $\mathbb{H}$ , which is simply connected; by Cauchy's theorem each integral is unchanged under any deformation fixing its endpoints, hence depends on those endpoints alone. This is path-independence.

For independence of  $\tau_0$  show that variations in  $\tau_0$  for the S-term and T-term in  $L^*(f, s)$  cancel, i.e. that  $\partial_{\tau_0} L_S^*(f, s; \gamma_{\tau_0}^S) = -\partial_{\tau_0} L_T^*(f, s; \gamma_{\tau_0}^T)$ . For  $L_T^*$ , note that the T-segment runs from  $T^{-1}\tau_0 = \tau_0 - 1$  to  $\tau_0$ , so by Leibniz's rule and the T-periodicity  $f(\tau_0 - 1) = f(\tau_0)$ ,

$$\partial_{\tau_0} L_T^*(f, s; \gamma_{\tau_0}^T) = f(\tau_0) \tilde{k}_T(\tau_0, s) - f(\tau_0 - 1) \tilde{k}_T(\tau_0 - 1, s) = f(\tau_0) \delta_T \tilde{k}_T(\tau_0 - 1, s), \quad (12)$$

where  $\delta_T \tilde{k}_T(\tau, s) := \tilde{k}_T(\tau + 1, s) - \tilde{k}_T(\tau, s)$ . This is the whole role of  $\delta_T$ : the unit gap between the endpoints turns the  $\partial_{\tau_0}$  boundary term into a unit shift of the kernel. That shift is elementary, since  $\zeta(\nu, x + 1) - \zeta(\nu, x) = -x^{-\nu}$  gives, on both terms at  $x = \tau + 1$ ,

$$\delta_T \tilde{k}_T(\tau, s) = -(\tau + 1)^{s-1} + e^{i\pi(s-1)} (\tau + 1)^{k-1-s} \quad (13)$$

$$\implies \partial_{\tau_0} L_T^*(f, s; \gamma_{\tau_0}^T) = f(\tau_0) [-\tau_0^{s-1} + e^{i\pi(s-1)} \tau_0^{k-1-s}]. \quad (14)$$

The S-segment runs  $-1/\tau_0 \rightarrow \tau_0$ ; its lower endpoint moves with  $\partial_{\tau_0}(-1/\tau_0) = \tau_0^{-2}$ , so by Leibniz, the modularity  $f(-1/\tau_0) = \tau_0^k f(\tau_0)$ , and the phase  $(-1/\tau_0)^{s-1} = e^{i\pi(s-1)} \tau_0^{1-s}$ ,

$$\partial_{\tau_0} L_S^*(f, s; \gamma_{\tau_0}^S) = f(\tau_0) \tau_0^{s-1} - f(-1/\tau_0) (-1/\tau_0)^{s-1} \tau_0^{-2} = f(\tau_0) [\tau_0^{s-1} - e^{i\pi(s-1)} \tau_0^{k-1-s}]. \quad (15)$$

Thus  $\tau_0$ -variations are equal and opposite,  $\partial_{\tau_0} L_S^* = -\partial_{\tau_0} L_T^*$ . Thus  $\partial_{\tau_0} L^* = \partial_{\tau_0} L_S^* + \partial_{\tau_0} L_T^* = 0$ .  $\square$

**Remark 2.7** (Recovering the classical L-integral for  $f \in S_k$ ). *Let  $f \in S_k$ . As  $f^{-1}(\infty)$  does not include the cusp  $i\infty$ ,  $\lim_{\tau_0 \rightarrow i\infty} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  exists. In this limit, the S-segment maps onto the standard L-function contour,  $[0, i\infty]$ . Simultaneously, as  $f(it) \sim e^{-2\pi t}$  as  $t \rightarrow \infty$ , the T-segment vanishes exponentially. Thus, in this limit,  $L^*(f, s)$  exactly maps onto the classical L-function integral [4]:*

$$\lim_{\tau_0 \rightarrow i\infty} L^*(f, s) = \int_0^\infty t^{s-1} f(it) dt. \quad (16)$$

### 3 L-functions for weak modular forms

**Proposition 3.1.** *Let  $\Gamma(s, x) := \int_x^\infty e^{-t} t^{s-1} dt$  denote the incomplete Gamma function. For  $n \neq 0$ ,  $\nu \in \mathbb{C}$ , and  $z \in \mathbb{H}$ ,*

$$\int_z^{z+1} e^{2\pi i n u} \zeta(\nu, u) du = e^{i\pi(1-\nu)/2} (2\pi n)^{\nu-1} \Gamma(1-\nu, -2\pi i n z). \quad (17)$$

*Proof.* Write  $\zeta(\nu, u) = \sum_{m \geq 0} (m+u)^{-\nu}$  and shift  $u \mapsto u+m$  in the  $m$ -th term; as  $e^{2\pi i n(u+m)} = e^{2\pi i n u}$  for integer  $m$ , the unit segments stack into one horizontal ray,

$$\int_z^{z+1} e^{2\pi i n u} \zeta(\nu, u) du = \sum_{m \geq 0} \int_{z+m}^{z+1+m} e^{2\pi i n v} v^{-\nu} dv = \int_z^{z+\infty} e^{2\pi i n v} v^{-\nu} dv. \quad (18)$$

Now, substitute  $w = -2\pi i n v$ , so that  $v = iw/(2\pi n)$ ,  $dv = i dw/(2\pi n)$ , and  $e^{2\pi i n v} = e^{-w}$ : the lower endpoint becomes  $w = -2\pi i n z$ , and for  $n > 0$  the ray rotates to  $w \rightarrow +\infty$  along the positive reals. Since  $e^{-w} w^{-\nu}$  decays in the right half-plane, the closing arc vanishes. With  $v^{-\nu} = i^{-\nu} (2\pi n)^{\nu} w^{-\nu}$ ,

$$\int_z^{z+\infty} e^{2\pi i n v} v^{-\nu} dv = i^{1-\nu} (2\pi n)^{\nu-1} \int_{-2\pi i n z}^{\infty} e^{-w} w^{-\nu} dw = e^{i\pi(1-\nu)/2} (2\pi n)^{\nu-1} \Gamma(1-\nu, -2\pi i n z). \quad (19)$$

For  $n < 0$  the ray instead grows as  $\text{Im} v \rightarrow \infty$  and the rotation is illegal; but both sides are analytic in  $n$  and agree for  $n > 0$ , so the identity continues to  $n < 0$  with  $\Gamma$  on its principal branch. At  $n = 0$  the prefactor  $(2\pi n)^{\nu-1}$  is singular; that case is handled separately.  $\square$

**Theorem 3.2** (The L-function for  $f \in M_k^!$ ). *For  $f \in M_k^!$ , we may set  $\tau_0 = i$ , and prove*

$$L^*(f, s) = -c_f(0) \left( \frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left( \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right). \quad (20)$$

Here, trivially,  $L^*(f, s) = i^k L^*(f, k-s)$ . The L-function is  $L(f, s) := (2\pi)^s L^*(f, s)/\Gamma(s)$ .

*Proof.* By path independence and base-point independence from Proposition 2.6, we may freely set  $\tau_0 = i$ . Here,  $L_S^*(f, s; \gamma_{\tau_0}^S) = 0$  (the  $S$ -contour vanishes in length and there are no nontrivial homotopy classes as  $f \in M_k^!$  has no poles which a finite contour can wrap), and the L-integral simplifies dramatically:

$$L^*(f, s) = L_T^*(f, s; \gamma_i^T) = e^{-\pi i s/2} \int_{i-1}^i f(\tau) \tilde{k}_T(\tau, s) d\tau. \quad (21)$$

We then expand  $f$  in terms of its q-series,  $\sum_n c_f(n) e^{2\pi i n \tau}$ , and substitute the  $u$ -variable of Proposition 3.1 as  $u = \tau + 1$ , and note that  $e^{2\pi i n(u-1)} = e^{2\pi i n u}$ . When  $n \neq 0$ , straightforward substitutions of  $\nu, z$  into Proposition 3.1 suffice:  $1-\nu = s$  gives the contribution of the first term in  $\tilde{k}_T(\tau, s)$ , and  $1-\nu = k-s$  gives the second:

$$e^{-\pi i s/2} \left( e^{\pi i s/2} \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} - e^{i\pi(s-1)} e^{\pi i(k-s)/2} \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right) = \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}}. \quad (22)$$

We may then evaluate the  $n = 0$  mode's contribution follows from noting the closed form solution for the definite integral,

$$\int_{i-1}^i \zeta(\sigma, \tau + 1) d\tau = \int_i^{i+1} \zeta(\sigma, u) du = \frac{i^{1-\sigma}}{\sigma-1}. \quad (23)$$

This result follows from noting  $\partial_a(\zeta(\sigma-1, a)/(1-\sigma)) = \zeta(\sigma, a)$  and  $\zeta(\sigma-1, a+1) - \zeta(\sigma-1, a) = -a^{1-\sigma}$ . Applying this to the two distinct terms of  $\tilde{k}_T(\tau, s)$  at  $\sigma = 1-s$  and  $\sigma = 1-(k-s)$  gives

$$e^{-\pi is/2} \int_{i-1}^i c_f(0) \tilde{k}_T(\tau, s) d\tau = -c_f(0) \left( \frac{1}{s} + \frac{i^k}{k-s} \right). \quad (24)$$

Combining the  $n=0$  contributions with the  $n \neq 0$  contributions reproduces Eq. (20). Finally, the functional equation,  $L^*(f, s) = i^k L^*(f, k-s)$ , follows by direct inspection.  $\square$

### 3.1 Explicit S- and T-contours for general $\tau_0$

**Lemma 3.3** ( $L_T^*$  for  $M_k^!$ ). *For  $f \in M_k^!$ , and  $\tau_0 \in \mathbb{H}$  away from cusps,  $L_T^*(f, s; \gamma_{\tau_0}^T)$  evaluates to:*

$$L_T^*(f, s; \gamma_{\tau_0}^T) = -c_f(0) \left( \frac{(\tau_0/i)^s}{s} + i^k \frac{(\tau_0/i)^{k-s}}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left( \frac{\Gamma(s, -2\pi in\tau_0)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, -2\pi in\tau_0)}{(2\pi n)^{k-s}} \right). \quad (25)$$

*Proof.* Expand  $f = \sum_n c_f(n) q^n$  on the compact segment  $[\tau_0-1, \tau_0]$ , and restrict attention to the sub-sum,  $\sum_{n \neq 0}$ . For each such  $n \neq 0$ , apply the Proposition 3.1 in  $\tilde{k}_T(\tau, s) = \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-(k-1), \tau+1)$  at  $\nu = 1-s$  and  $\nu = s-(k-1)$ . The phases for the second term work-out identically as in the proof of Theorem 3.2.

For  $n=0$ , the constant  $c_f(0)$  multiplies  $e^{-\pi is/2} \int_{\tau_0-1}^{\tau_0} \tilde{k}_T(\tau, s) d\tau$ , which we compute as follows. First, the closed-form of the Hurwitz antiderivative  $\int_{\tau_0-1}^{\tau_0} \zeta(\sigma, \tau+1) d\tau = \tau_0^{1-\sigma}/(\sigma-1)$  comes from  $\partial_\tau \zeta(\sigma-1, \tau+1) = -(\sigma-1)\zeta(\sigma, \tau+1)$  and the shift identity  $\zeta(\sigma-1, \tau_0+1) - \zeta(\sigma-1, \tau_0) = -\tau_0^{1-\sigma}$ . Finally, note that the two terms in  $\tilde{k}_T$  are simply specializations of  $\sigma = 1-s$  and  $\sigma = s-k+1$ , which yield  $-e^{-\pi is/2} \tau_0^s/s = -(\tau_0/i)^s/s$  and  $-e^{\pi is/2} \tau_0^{k-s}/(k-s) = -i^k (\tau_0/i)^{k-s}/(k-s)$  respectively.  $\square$

**Corollary 3.4** ( $L_T^*$  as  $\tau_0 \rightarrow i\infty$ ). *Let  $f \in S_k^!$ , where  $f = \sum_{n \geq -N} c_f(n) q^n$  for some  $-N \in \mathbb{Z}$ . The T-contour's contribution to  $L^*(f, s)$  is given by Lemma 3.3. As  $\tau_0 = it$  goes to the cusp  $i\infty$ , the T-contour scales exactly as  $f(it)$  scales:  $L_T^*(f, s; \gamma_{\tau_0}^T) \sim f(it) \sim e^{2\pi Nt}$ .*

*Proof.* For the asymptotics,  $\Gamma(\sigma, z) \sim z^{\sigma-1} e^{-z}$  as  $|z| \rightarrow \infty$  for  $|\arg z| < \frac{3\pi}{2}$  [10, §8.11] gives  $|c_f(n)|$ -weighted go as  $e^{-2\pi n \text{Im}(\tau_0)}$ . The smallest mode  $n$  in the q-series, i.e.  $n = -N$ , dominates the scaling of  $f(it)$  and of  $L_T^*(f, s; \gamma_{it}^T)$ . Combining these results yields the claimed asymptotics.  $\square$

Now, exploiting  $\tau_0$ -independence of  $L^*(f, s)$ , its explicit value as given in Theorem 3.2, and the explicit T-contribution  $L_T^*(f, s; \gamma_{\tau_0}^T)$  trivially implies clean, closed-form expressions for  $L_S^*(f, s; \gamma_{\tau_0}^S)$ :

**Corollary 3.5** ( $L_S^*$  for  $M_k^!$ ). *For  $f \in M_k^!$  and  $\tau_0 \in \mathbb{H}$ ,  $L_S^*(f, s; \gamma_{\tau_0}^S)$  evaluates to*

$$L_S^*(f, s; \gamma_{\tau_0}^S) = c_f(0) \left( \frac{(\tau_0/i)^s - 1}{s} + i^k \frac{(\tau_0/i)^{k-s} - 1}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left( \frac{\Gamma(s, 2\pi n) - \Gamma(s, -2\pi in\tau_0)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n) - \Gamma(k-s, -2\pi in\tau_0)}{(2\pi n)^{k-s}} \right). \quad (26)$$

*Proof.* Recall  $L^*(f, s) - L_T^*(f, s; \gamma_{\tau_0}^T) = L_S^*(f, s; \gamma_{\tau_0}^S)$ . Now,  $L^*(f, s)$  for  $f \in M_k^!$  is given by Theorem 3.2, and  $L_T^*(f, s; \gamma_{\tau_0}^T)$  is given by Lemma 3.3. Their difference yields Eq. (26).  $\square$

### 3.2 Classical Mellin recovery and canonical uniqueness.

As discussed in Remark 2.7, for  $f \in S_k$ , taking the cusp-limit  $\tau_0 \rightarrow i\infty$  in Theorem 3.2 collapses the  $S$ -segment onto the classical Mellin contour  $[0, i\infty]$  while via Corollary 3.4 the  $T$ -segment vanishes exponentially as  $L_T^*(f, s; \gamma_{\tau_0}^T) \sim f(\tau_0) \sim e^{-2\pi\text{Im}(\tau_0)N}$  (as  $f \in S_k \Leftrightarrow f = \sum_{n \geq N} c_f(n)q^n$  with  $N \geq 1$ ). I.e., here, one recovers the classical L-integral of  $f$  without obstruction:

$$\lim_{\tau_0 \rightarrow i\infty} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = \int_0^\infty t^{s-1} f(it) dt. \quad (27)$$

This story changes when  $f = 1/q^N(1 + O(q)) \in S_k^!$  for some  $N \geq 1$ .

In short, when  $f$  has a polar  $q$ -series, it has a pole at the cusp of  $\text{SL}_2(\mathbb{Z})$ . Thus, the limit  $\tau_0 \rightarrow i\infty$ , which was crucial to recovering the classical Mellin contour in Remark 2.7 simply does not exist.

However, it can still be instructive to examine  $L_T^*(f, s; \gamma_{\tau_0}^T)$ , and  $L_S^*(f, s; \gamma_{\tau_0}^S)$  as  $\tau_0$  goes towards  $i\infty$ . In this context, while of course the sum  $L_T^*(f, s; \gamma_{\tau_0}^T) + L_S^*(f, s; \gamma_{\tau_0}^S) = L^*(f, s)$  is manifestly invariant under changes of the base-point, each of these two terms can indeed change values. Indeed, Corollary 3.4 clearly shows that as  $\tau_0 \rightarrow i\infty$ , when  $f \in S_k^!$  has negative powers of  $q$  in its  $q$ -series, then  $L_T^*(f, s; \gamma_{\tau_0}^T) \sim e^{2\pi N \text{Im}(\tau_0)}$ . Discarding  $\gamma_{\tau_0}^T$  therefore does violence to the result: When the  $S$ -contour is isolated from its  $T$ -partner, and  $\tau_0 \rightarrow i\infty$ , it diverges. This is the precise reason that [1] needed to regularize their divergent L-integral representation for properly weak cusp forms.

That said, there is one further aspect of our L-function even in the classical case of  $f \in M_k$  that is worthy of remark:

**Remark 3.6** (Recovering the classical L-integral for  $f \in M_k$ ). *In textbook treatments [4], when  $f \in M_k$ , its L-integral is defined as*

$$\begin{aligned} L^*(f, s)|_{\text{cl}} &:= \int_0^\infty (f(it) - c_f(0))t^{s-1}dt = \frac{\Gamma(s)}{(2\pi)^s} \sum_{n=1}^\infty \frac{c_f(n)}{n^s} \\ &= -c_f(0) \left( \frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n=1}^\infty c_f(n) \left( \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right) \end{aligned} \quad (28)$$

*Classically, this constant-term subtraction is needed because  $\int_0^\infty t^{s-1}dt$  is a pure divergence, so the  $q$ -series cannot commute with the integration contour.*

*However, as is clear from our general Theorem 3.2 for L-integrals for  $f \in M_k^!$ , and from the specific results of the  $S$ - and  $T$ -contours in Lemma 3.3 and Corollary 3.5, our L-integral requires no such explicit subtraction of the constant term  $-c_f(0)$ . Thus, not only does our prescription naturally handle divergences at poles when  $f \in M_k^!$  has polar terms in its  $q$ -series, but it also obviates the need to by-hand remove the constant mode from the kernel of the L-integral that is needed when defining L-functions in terms of the classical Mellin contour,  $\int_0^\infty$ .*

### 3.3 Period polynomials, periods and quasi-periods

Because  $L^*(f, s)$  is canonical for every  $f \in S_k^!$ , and because period polynomials for a given modular form can be written in terms of their L-functions evaluated at integer- $s$  in the critical strip, they need no separate construction.

**Definition 3.7** (Period polynomial for  $S_k^!$ ). *For  $f \in S_k^!$  and  $n = k - 2$ , the period polynomial is*

$$r_f(X, Y) := (2\pi i)^{n+1} \sum_{\ell=0}^n (-1)^\ell \binom{n}{\ell} i^{\ell+1} L^*(f, \ell+1) X^{n-\ell} Y^\ell. \quad (29)$$

**Proposition 3.8** (Period polynomial as a contour integral). *Let  $n := k - 2$ . With  $(X - \tau Y)^n$  in place of the  $L$ -weight  $\tau^{s-1}$ , and  $\mathbf{K}_T(\tau; X, Y) := \sum_{\ell=0}^n \binom{n}{\ell} (-1)^\ell \tilde{k}_T(\tau, \ell + 1) X^{n-\ell} Y^\ell$  in place of  $\tilde{k}_T$ , the same two segments assemble  $r_f$  directly:*

$$r_f(X, Y) = (2\pi i)^{n+1} \left[ \int_{\gamma_{\tau_0}^S} f(\tau) (X - \tau Y)^n d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \mathbf{K}_T(\tau; X, Y) d\tau \right]. \quad (30)$$

*Proof.* By Definition 2.3, the  $X^{n-\ell} Y^\ell$  coefficient of the bracket is

$$r_f|_{X^{n-\ell} Y^\ell} = (-1)^\ell \binom{n}{\ell} i^{\ell+1} [L_S^*(f, \ell + 1) + L_T^*(f, \ell + 1)] = (-1)^\ell \binom{n}{\ell} i^{\ell+1} L^*(f, \ell + 1). \quad (31)$$

This matches Definition 3.7 term by term, and proves the claim.  $\square$

Having now defined the period polynomial associated to a weak cusp form  $f \in S_k^!$ , we now seek to find its (*quasi*-)periods. First, a number of helpful definitions. For starters, let  $f \in S_k^!$ ,  $\tau_0 \in \mathbb{H}$  and  $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ , and define

$$C_\gamma(f; \tau_0) := (2\pi i)^{n+1} \int_{\gamma^{-1}\tau_0}^{\tau_0} \omega_f, \quad \text{where } \omega_f := f(\tau)(X - \tau Y)^n d\tau. \quad (32)$$

Finally, it is helpful to define two slight abstractions related to  $\mathbf{K}_T(\tau; X, Y)$  and to  $L_T^*(f, s; \gamma_{\tau_0}^T)$ :

$$\chi(\tau; X, Y) := \sum_{\ell=0}^n \binom{n}{\ell} (-1)^\ell \zeta(-\ell, \tau + 1) X^{n-\ell} Y^\ell, \quad (33)$$

$$E(f; \tau_0) := (2\pi i)^{n+1} \int_{\gamma_{\tau_0}^T} f(\tau) \chi(\tau; X, Y) d\tau. \quad (34)$$

For each  $\tau$ ,  $\chi(\tau; \cdot, \cdot) \in V_{k-2} = \mathbb{C}[X, Y]_{k-2}$ ; the  $\mathrm{SL}_2(\mathbb{Z})$ -slash  $(P|_\gamma)(X, Y) := P(aX + bY, cX + dY)$  acts on the polynomial variables alone, holding  $\tau$  fixed:

$$\chi|_\gamma(\tau; X, Y) := \chi(\tau; aX + bY, cX + dY), \quad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}). \quad (35)$$

Thus  $|_T$  sends  $(X, Y) \mapsto (X + Y, Y)$  with  $\tau$  untouched. Being linear and  $\tau$ -independent,  $|_\gamma$  commutes with  $\int d\tau$ , so slashing the polynomial  $E$  agrees with slashing  $\chi$  under the integral; the two occurrences of  $|_\gamma$  in Lemma 3.9 denote the same operation.

**Lemma 3.9.** *Let  $f \in S_k^!$ ,  $\tau_0 \in \mathbb{H}$ , and let  $\omega_f$ ,  $C_\gamma(f; \tau_0)$ ,  $\chi(\tau; X, Y)$ , and  $E(f; \tau_0)$  be as defined in Eqs. (32), (33) and (34). Then  $\hat{C}_\gamma := C_\gamma - (E|_\gamma - E)$  is a 1-cocycle for  $\mathrm{SL}_2(\mathbb{Z})$  with  $\hat{C}_S = r_f$  and  $\hat{C}_T = 0$ .*

*Proof.* First, we prove  $\hat{C}_\gamma$  is a cocycle. Path-additivity and covariance of  $\omega_f$  give the Eichler–Shimura cocycle relation  $C_{\gamma\delta} = C_\gamma|_\delta + C_\delta$  [11, 6]; as  $\gamma \mapsto E|_\gamma - E$  is a coboundary,  $\hat{C}$  is a cocycle.

Having established it is a cocycle, we now show that  $\hat{C}_S = r_f$ . At  $s = \ell + 1$ , Definition 2.2 reads  $\tilde{k}_T(\tau, \ell + 1) = \zeta(-\ell, \tau + 1) - (-1)^\ell \zeta(-(n - \ell), \tau + 1)$ ; assembling  $\mathbf{K}_T$  and re-indexing  $\ell \mapsto n - \ell$  in the second term (with  $n$  even) gives  $\mathbf{K}_T = \chi - \chi|_S$ . So the  $T$ -term of  $r_f$  is  $E - E|_S$ , and by Proposition 3.8  $r_f = C_S + (E - E|_S) = C_S - (E|_S - E) = \hat{C}_S$ .

Finally, we show that  $\hat{C}_T = 0$ . The Hurwitz shift  $\zeta(-\ell, \tau) = \zeta(-\ell, \tau + 1) + \tau^\ell$  gives the point-wise identity  $\chi(\tau - 1; X, Y) - \chi(\tau; X, Y) = (X - \tau Y)^n$ ; applied twice it shows  $A(\tau) :=$

$\chi(\tau; X+Y, Y) - \chi(\tau-1; X, Y)$  satisfies  $A(\tau-1) = A(\tau)$ . Crucially, because  $A(\tau)$  is a polynomial, and because  $A(\tau) = A(\tau+1)$  for generic  $\tau$ , then  $A(\tau)$  is constant:  $\partial_\tau A(\tau) = 0 \Leftrightarrow A(\tau) = A_0$ . Thus,

$$\begin{aligned}
\hat{C}_T &= C_T - (E|_T - E) \\
&= (2\pi i)^{n+1} \int_{\gamma_{\tau_0}^T} f(\tau)(X - \tau Y)^n d\tau - (2\pi i)^{n+1} \int_{\gamma_{\tau_0}^T} f(\tau) [\chi|_T(\tau; X, Y) - \chi(\tau; X, Y)] d\tau \\
&= (2\pi i)^{n+1} \int_{\gamma_{\tau_0}^T} f(\tau) \left( [\chi(\tau-1; X, Y) - \chi(\tau; X, Y)] - [\chi(\tau; X+Y, Y) - \chi(\tau; X, Y)] \right) d\tau \\
&= (2\pi i)^{n+1} \int_{\gamma_{\tau_0}^T} f(\tau) [\chi(\tau-1; X, Y) - \chi(\tau; X+Y, Y)] d\tau = -(2\pi i)^{n+1} \int_{\tau_0-1}^{\tau_0} f(\tau) A(\tau) d\tau \\
&= -(2\pi i)^{n+1} A_0 \int_{\tau_0-1}^{\tau_0} f(\tau) d\tau = -(2\pi i)^{n+1} A_0 c_f(0) = 0,
\end{aligned} \tag{36}$$

as all elements  $f \in S_k^!$  must have  $c_f(0) = 0$ .  $\square$

**Definition 3.10.**  $W := \{w \in \mathbb{C}[X, Y]_{k-2} : w|_{(1+S)} = w|_{(1+U+U^2)} = 0\} \subset V_n$

**Lemma 3.11.** For  $f \in S_k^!$ , and  $W \subset V_n$  from Definition 3.10, the period polynomial  $r_f(X, Y) \in W$ .

*Proof.* Let  $\hat{C}_\gamma$  be the parabolic cocycle of Lemma 3.9. Thus,  $r_f = \hat{C}_S$  and  $\hat{C}_T = 0$ . Now, note that  $\hat{C}_\gamma(f; \tau_0) = 0$  when  $\gamma = \pm I$  for the simple reason that when  $\gamma = \pm I$ ,  $\hat{C}_\gamma$  is a closed-contour integral of a function which has no poles in  $\mathbb{H}$ . Finally, direct computation, starting from the identities  $-I = U^3 = S^2$  and using the cocycle law  $\hat{C}_{\gamma\delta} = \hat{C}_\gamma|_\delta + \hat{C}_\delta$  and clever use of  $\hat{C}_T = 0$ , shows  $r_f|_{(1+S)} = r_f|_{(1+U+U^2)} = 0$ :

$$\hat{C}_U = \hat{C}_{TS} = \hat{C}_T|_S + \hat{C}_S = \hat{C}_S = r_f, \tag{37}$$

$$0 = \hat{C}_{S^2} = \hat{C}_S|_S + \hat{C}_S = \hat{C}_S|_{(1+S)} = r_f|_{(1+S)}, \tag{38}$$

$$0 = \hat{C}_{U^3} = \hat{C}_U|_{UU} + \hat{C}_{UU} = \hat{C}_U|_{U^2} + (\hat{C}_U|_U + \hat{C}_U) = \hat{C}_U|_{(1+U+U^2)} = r_f|_{(1+U+U^2)}. \tag{39}$$

Thus,  $r_f \in W \subset V_n$ , and finishes the proof.  $\square$

To define periods and quasi-periods, we recall the standard fact that  $\dim(W) = 2 \dim(S_k) + 1$  (see, e.g. [6]), and define the inner-product:

$$\langle P, Q \rangle := \sum_a (-1)^a \binom{n}{a}^{-1} p_a q_{n-a}. \tag{40}$$

Now for the basis elements for  $k$  where  $\dim(S_k) = 1$ :

**Definition 3.12.** Let  $\dim S_k = 1$ ,  $n := k-2$ ,  $p_0(X) := X^n - 1$ ,  $B_\ell := \frac{\ell!}{2\pi i} \oint \frac{t^{-\ell}}{e^t - 1} dt$ , and  $B_m^0(X) := \sum_{i \neq 1} \binom{m}{i} B_i X^{m-i}$ . Motivated by [6, Thm. 1'], we define:

$$W_-^{(k)}(X) := \text{pp} \left( \frac{1}{n-1} (B_{n-1}^0(X) + X^n B_{n-1}^0(\frac{1}{X})) - \frac{1}{3} (B_3^0(X) + X^n B_3^0(\frac{1}{X})) \right), \tag{41}$$

$$W_+^{(k)}(X) := \text{pp} \left( \frac{1}{2} (B_2^0(X) - X^n B_2^0(\frac{1}{X})) + \frac{1}{n} (B_n^0(X) - X^n B_n^0(\frac{1}{X})) - c p_0(X) \right), \tag{42}$$

$$v_0^{(k)}(X) := p_0(X) - W_+^{(k)}(X) \left( \langle W_+^{(k)}, p_0 \rangle / \langle W_+^{(k)}, W_+^{(k)} \rangle \right), \tag{43}$$

where  $\text{pp}$  acts on any polynomial with rational coefficients by multiplying by all coefficient-denominators and dividing by the gcd,  $c$  eliminates the  $p_0(X)$  component of  $W_+^{(k)}$ , and  $\langle P, Q \rangle$  as in Eq. (40).

**Lemma 3.13.** *For  $k$  with  $\dim S_k = 1$ , the space  $W$  of Lemma 3.11 has  $\dim W = 3$ , and  $\{W_+^{(k)}, W_-^{(k)}, v_0^{(k)}\}$  is an orthogonal basis of  $W$  under  $\langle \cdot, \cdot \rangle$  from Eq. (40).*

*Proof.* First, let  $W_\pm^{(k)}(X, Y) := Y^{k-2}W_\pm^{(k)}(X/Y)$ , and let  $v_0^{(k)}(X, Y) := Y^{k-2}v_0^{(k)}(X/Y)$ .

Now, by Eichler–Shimura,  $\dim W = 2 \dim S_k + 1 = 3$ . Theorem 1' of [6] gives the odd and even period polynomials of the cusp form  $R_n \propto \Delta$  as modified-Bernoulli combinations; its  $n = 2$  case is  $r^-(R_2)$ , odd under  $X \mapsto -X$ , and its  $n = 1$  case is  $r^+(R_1)$ , even. Our  $W_-^{(k)}$  is the primitive part of  $r^-(R_2)$ , and  $W_+^{(k)}$  the primitive part of  $r^+(R_1)$  with its Eisenstein  $p_0$ -component removed by  $c$ ; both lie in  $W$ , as does  $p_0 = X^{k-2} - Y^{k-2}$  (even). The three are independent, hence a basis of  $W$ . Since  $k - 2$  is even,  $\langle \cdot, \cdot \rangle$  is symmetric and pairs opposite  $X \mapsto -X$  parities to zero; the odd  $W_-^{(k)}$  is thus orthogonal to the even  $W_+^{(k)}$  and  $v_0^{(k)}$ , while  $v_0^{(k)} \perp W_+^{(k)}$  by construction. Hence  $\{W_+^{(k)}, W_-^{(k)}, v_0^{(k)}\}$  is orthogonal.  $\square$

**Definition 3.14** (Periods and Quasi-Periods). *Let  $k$  be such that  $\dim(S_k) = 1$ ,  $n := k - 2$ ,  $W_\pm^{(k)}(X, Y) := Y^n W_\pm^{(k)}(X/Y)$  be as in Definition 3.12, and  $r_f(X, Y)$  be as in Definition 3.7. Now, define the two numbers  $\omega^\pm(f)$  as follows:*

$$\omega^\pm(f) := \frac{\langle r_f, W_\pm^{(k)} \rangle}{\langle W_\pm^{(k)}, W_\pm^{(k)} \rangle}, \quad (44)$$

where  $\langle \cdot, \cdot \rangle$  is as in Eq. (40). When  $f$  is a cusp form,  $\omega^\pm(f)$  are its periods; when  $f$  is a weak cusp form,  $\omega^\pm(f)$  are its quasi-periods, sometimes denoted  $\eta^\pm(f)$ .

Now, having defined our periods and quasi-periods, we use them together with our L-functions to extract periods and quasi-periods for  $f \in S_k^!$  (when  $\dim(S_k) = 1$ ). To this effect, we recall the notation of [5]. Decomposing the modular weight  $k = 12\lambda + k'$  where  $k' \in \{0, 4, 6, 8, 10, 14\}$ , we see  $\lambda = \dim(S_k)$  when  $\lambda \geq 0$ . For any  $\lambda$ , we may denote  $f_{k,m} := 1/q^m + O(q^{\lambda+1}) \in M_k^!$ . When  $\dim(S_k) = 1$ , we have  $f_{k,m} = 1/q^m + O(q^2)$ . Here, define  $\Delta_k := f_{k,-1} = q + O(q^2) \in S_k$  and  $\hat{\Delta}_k := f_{k,1} = 1/q + O(q^2) \in S_k^!$ . Further, define

$$\text{pd}_k := \begin{pmatrix} \omega^+(\Delta_k) & \omega^-(\Delta_k) \\ \eta^+(\hat{\Delta}_k) & \eta^-(\hat{\Delta}_k) \end{pmatrix}, \quad \alpha_k := \frac{(-1)^{1+k/2} \det(\text{pd}_k)}{(2\pi i)^{k-1} (k-2)!} \quad (45)$$

| $k$ | $\omega^+(\Delta_k)$ | $\omega^-(\Delta_k)/i$ | $\eta^+(\hat{\Delta}_k)$ | $\eta^-(\hat{\Delta}_k)/i$ | $\alpha_k$ |
|-----|----------------------|------------------------|--------------------------|----------------------------|------------|
| 12  | -6.89168e7           | -5.58502e6             | 1.27202e11               | 1.02767e10                 | 1          |
| 16  | -8.81481e11          | -1.54063e10            | 4.14120e13               | 6.83541e11                 | 13/30      |
| 18  | -5.07833e13          | -5.43305e12            | -2.19200e14              | -3.36339e13                | 2/3        |
| 20  | -4.22882e16          | -3.73170e14            | 6.38640e16               | -1.89823e14                | 17/5       |
| 22  | -1.49974e18          | -4.24489e16            | 1.43795e18               | -4.41050e16                | 19/21      |
| 26  | -1.18673e22          | -3.60961e21            | 1.18653e22               | -3.61016e21                | 23/15      |

Table 1: Periods  $\omega^\pm(\Delta_k)$  and quasi-periods  $\eta^\pm(\hat{\Delta}_k)$ , and the rational number  $\alpha_k$ . While  $\alpha_{12} = 1$  matches Brown [2],  $\alpha_k$  for  $k \geq 16$  appear to be new, interesting, rational numbers, albeit with implicit dependence on the period-polynomial basis normalizations—in contrast to the normalization-free Petersson check above. We have not found a clear pattern for the  $\alpha_k$ . See [9] for the scripts that extracted these figures (to 50-digit precision).

Brown–Hain [7, Prop. 5.6] show that  $\det(\text{pd}_k)$  satisfies a Legendre-type relation,  $\det(\text{pd}_k) \in \mathbb{Q}^\times \cdot (2\pi i)^{k-1}$ . Brown [2, §8.3] numerically finds  $\alpha_k = 1$ ; we reproduce this result at  $k = 12$  and

give the analogous rational multiples for  $k \in \{16, 18, 20, 22, 26\}$  in Table 1, below. Note: The value of  $\alpha_k$  is tied to the normalization of Definition 3.12. Other normalizations, e.g. not including the pp-operator in defining  $W_{\pm}^{(k)}$  would lead to different  $\alpha_k$ -values.

## 4 Meromorphic modular forms

For  $f \in F_k$  with poles in  $\mathbb{H}$ , the two segment integrals of Definition 2.3 depend on their contours; we show their sum depends only on the homotopy class of the pair.

**Proposition 4.1.** *For  $f \in F_k$  and  $\tau_0 \in \mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ , the L-integral  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  depends only on the homotopy class of its two segments in  $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ , and not on the basepoint  $\tau_0$ . At fixed  $\tau_0$  each of  $L_S^*, L_T^*$  is separately path-independent.*

*Proof.* At fixed  $\tau_0$  the integrands  $f(\tau)\tau^{s-1}$  and  $f(\tau)\tilde{k}_T(\tau, s)$  are holomorphic on  $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ , so Cauchy's theorem makes each of  $L_S^*, L_T^*$  constant on the homotopy class of its own segment. Under a motion of the basepoint,  $\partial_{\tau_0} L^*$  is evaluated at the moving endpoints alone; the computation of Proposition 2.6, which uses only  $f(\tau_0 - 1) = f(\tau_0)$  and  $f(-1/\tau_0) = \tau_0^k f(\tau_0)$ , cancels the boundary terms of  $L_S^*$  against those of  $L_T^*$ , the interior poles lying off the contours. Hence  $\partial_{\tau_0} L^* = 0$  along any motion in  $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ , while each summand may separately vary.  $\square$

**Definition 4.2.** *For  $f \in F_k$  and  $\tau_0 \in \mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ , let  $[\gamma_{\tau_0}^S], [\gamma_{\tau_0}^T]$  denote the homotopy classes of the two segments in  $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ .*

Thus, with  $\gamma_{\tau_0}^S \in [\gamma_{\tau_0}^S]$  and  $\gamma_{\tau_0}^T \in [\gamma_{\tau_0}^T]$ , by Proposition 4.1 the value

$$L^*(f, s; [\gamma_{\tau_0}^S], [\gamma_{\tau_0}^T]) := L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = L_S^*(f, s; \gamma_{\tau_0}^S) + L_T^*(f, s; \gamma_{\tau_0}^T) \quad (46)$$

is independent of the chosen representative paths and independent of  $\tau_0$  (subject to natural constraints of  $\tau_0$  monodromies around singular points, which move between homotopy classes).

We would like to speak of “the” L-integral. Homotopy classes of course split this into L-integrals for contours in different homotopy classes. And, so, in what follows, we define the reference homotopy classes:

**Definition 4.3** (Reference homotopy class for  $f \in F_k$ ). *Let  $f \in F_k$ , let  $\tau_0$  be the base-point of  $L^*(f, s)$ , and let  $f^{-1}(\infty)$  be the poles of  $f$ . If  $f$  has no poles on the line  $\text{Im}(\tau) = 1$ , choose  $\delta = 0$ . Else, choose  $\delta > 0$  such that no pole lies in the strip  $1 < \text{Im}(\tau) \leq 1 + \delta$ . Set  $\hat{\tau}_0^f := i(1 + \delta)$ . The reference homotopy class  $[\hat{\gamma}_f] = [\hat{\gamma}_f^T] + [\hat{\gamma}_f^S]$  is:*

- $[\hat{\gamma}_f^T]$ : contains the horizontal segment from  $T^{-1}\hat{\tau}_0^f$  to  $\hat{\tau}_0^f$ ;
- $[\hat{\gamma}_f^S]$ : contains the near-vertical path from  $S^{-1}\hat{\tau}_0^f$  to  $\hat{\tau}_0^f$  along the imaginary axis, taking a pole at  $i$  by principal value.

For  $\delta = 0$ , the reference paths match the paths  $\gamma_i^S : i \rightarrow i$  and  $\gamma_i^T : (i-1) \rightarrow i$  for  $L^*(f, s)$  when  $f \in M_k^1$ .

With this definition, we may thus unambiguously write

$$L^*(f, s) := L^*(f, s; [\hat{\gamma}_f^S], [\hat{\gamma}_f^T]) . \quad (47)$$

Now, having fixed the reference value  $L^*(f, s)$  in the reference homotopy class of Definition 4.3, consider two paths  $\gamma_{\tau_0}^S$  and  $\gamma_{\tau_0}^T$  which may be in any homotopy class. Further, let  $X_S(p), X_T(p) \in \mathbb{Z}$  count the windings  $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$  has about each pole  $\tau_p \in f^{-1}(\infty)$  relative to the reference-class.

**Lemma 4.4.** *Let  $f \in F_k$ ,  $r_S(f, \tau_p) := \text{Res}_{\tau=\tau_p}(f(\tau)\tau^{s-1})$  and  $r_T(f, \tau_p) := \text{Res}_{\tau=\tau_p}(f(\tau)\tilde{k}_T(\tau, s))$ . Then,*

$$L^*(f, s; [\gamma_{\tau_0}^S], [\gamma_{\tau_0}^T]) = L^*(f, s) + 2\pi i e^{-\pi i s/2} \sum_{\tau_p \in f^{-1}(\infty)} \left[ X_S(\tau_p) r_S(f, \tau_p) + X_T(\tau_p) r_T(f, \tau_p) \right]. \quad (48)$$

*Proof.* Two pairs whose winding data differ by one unit at a single pole  $p$ , on one segment, bound a loop encircling  $p$  once; the residue theorem adds  $2\pi i$  times the residue of that segment's integrand,  $f(\tau)\tau^{s-1}$  for  $\gamma^S$  or  $f(\tau)\tilde{k}_T(\tau, s)$  for  $\gamma^T$ , and the prefactor  $e^{-\pi i s/2}$  of Definition 2.3 multiplies through. Summing over crossings gives (48).  $\square$

This content provides the basics to discuss our finite L-functions for  $f \in F_k$ . However, we wish to present these L-functions in terms of their q-series coefficients, just as we have done for L-functions for  $f \in M_k^!$  in the previous section. To do this, we must introduce polylogarithmic projections of Ref. [3]:

This polylog projection isolates the q-series contribution of a given pole of an  $f \in F_k$  into a known special function, an integral of a polylogarithm of negative index, and leaves behind an object  $(I - \hat{P})f$  with that one pole removed. For our purposes, we find it essential to project-out all poles at, and above, the line  $\text{Im } \tau = 1$ . We must do this because the reference contours which describe our reference homotopy classes are, at a minimum, contour integrals of  $f$  across the horizontal segment from  $i - 1$  to  $i$ . Were we to allow poles at/above this line, q-series diverges on the contour ( $|c_f(n)| \gtrsim n^{m-1} e^{2\pi n \text{Im } \tau_p}$  for a pole of order  $m$  [3]).

Within this general context, we lay-out the starting definitions and facts:

**Definition 4.5.** *Let  $f \in F_k$  have, at a pole  $\tau_p$  of order  $m_p$ , the Laurent principal part  $\sum_{m=1}^{m_p} r_{f, \tau_p}^*(m) (\tau - \tau_p)^{-m}$ ; the pole weights  $r_{f, \tau_p}^*(m)$  are thus its Laurent coefficients. With the polylogarithm  $\text{Li}_s(x) := \sum_{n \geq 1} x^n n^{-s}$ , the polylog projection of  $f$  at  $\tau_p$  is*

$$\hat{P}_{\tau_p} f(\tau) := \sum_{m=1}^{m_p} r_{f, \tau_p}^*(m) \frac{(-2\pi i)^m}{(m-1)!} \text{Li}_{1-m}(e^{2\pi i(\tau - \tau_p)}), \quad (49)$$

and  $\hat{P} := \sum_{\text{Im } \tau_p \geq 1} \hat{P}_{\tau_p}$ .

**Lemma 4.6.** *The normalized block  $\frac{(-2\pi i)^m}{(m-1)!} \text{Li}_{1-m}(e^{2\pi i(\tau - \tau_p)})$  has principal part exactly  $(\tau - \tau_p)^{-m}$ , so  $\hat{P}_{\tau_p} f$  matches the principal part of  $f$  and  $(I - \hat{P}_{\tau_p})f$  is holomorphic at  $\tau_p$ .*

*Proof.* By the small- $t$  expansion  $\text{Li}_{1-m}(e^{-t}) = (m-1)! t^{-m} + O(1)$  with  $t = -2\pi i(\tau - \tau_p)$ , the block  $\frac{(-2\pi i)^m}{(m-1)!} \text{Li}_{1-m}(e^{2\pi i(\tau - \tau_p)})$  has a pure pole  $(\tau - \tau_p)^{-m} + O(1)$  at  $\tau_p$ , with no lower-order poles. Summing against the Laurent coefficients  $r_{f, \tau_p}^*(m)$  therefore reproduces the principal part of  $f$  term by term, leaving  $(I - \hat{P}_{\tau_p})f$  regular at  $\tau_p$ .  $\square$

**Definition 4.7.** *With  $\hat{P}$  as in Definition 4.5, let  $c_{Rf}(n)$  be the Fourier coefficients of the pole-projected form  $(I - \hat{P})f$  in its expansion valid for large  $\text{Im } \tau$ :*

$$(I - \hat{P})f(\tau) = \sum_n c_{Rf}(n) q^n, \quad q = e^{2\pi i \tau}. \quad (50)$$

**Definition 4.8.**  *$F(\infty)$  is the space of meromorphic functions  $g : \mathbb{H} \rightarrow \mathbb{C}$  with  $g(\tau + 1) = g(\tau)$ , finitely many poles in each compact subset of the strip  $|\text{Re } \tau| \leq \frac{1}{2}$ , and bounded pole order at  $q = 0$ .*

**Lemma 4.9.**  $F_k \subset F(\infty)$  and  $(I - \hat{P})f \in F(\infty)$ . Every pole of  $(I - \hat{P})f$  has  $\text{Im } \tau < 1$ , with a maximal height  $y_b < 1$  (set  $y_b = 0$  if  $(I - \hat{P})f$  is holomorphic).

*Proof.* A weakly holomorphic form is  $T$ -periodic, meromorphic, and  $O(e^{C\text{Im}\tau})$  at the cusp, so  $F_k \subset F(\infty)$ ; the projected polylogs  $\text{Li}_{1-m}(e^{2\pi i(\tau-\tau_p)})$  share these properties, so  $(I - \hat{P})f \in F(\infty)$  (and leaves  $F_k$ , being non-modular). By Lemma 4.6,  $\hat{P}$  cancels the principal part of every pole with  $\text{Im}\tau \geq 1$ , so the poles of  $(I - \hat{P})f$  lie strictly below  $\text{Im}\tau = 1$ ; finite in each compact set and accumulating only at  $\text{Im}\tau = 0$ , they have a maximal height  $y_b < 1$ .  $\square$

The Fourier coefficients of a pole-projected form are controlled by its highest pole, an estimate we import from [3]:

**Lemma 4.10.** With  $y_b$  as in Lemma 4.9 and  $m_b$  the maximal order among poles at height  $y_b$ , there is a  $C > 0$  such that, for every  $\varepsilon > 0$ ,

$$C n^{m_b-1} e^{2\pi n y_b} \leq |c_{Rf}(n)| \leq e^{2\pi n(y_b+\varepsilon)}, \quad (51)$$

the upper bound for all large  $n$ , the lower for infinitely many  $n$ . In particular  $|c_{Rf}(n)| e^{-2\pi n} \rightarrow 0$  geometrically, at base  $e^{-2\pi(1-y_b-\varepsilon)}$ .

*Proof.* This is the pole-height/order Fourier-growth estimate of [3] for a  $T$ -periodic meromorphic function, applied to  $(I - \hat{P})f$  (Lemma 4.9).  $\square$

For  $n \neq 0$ , define

$$G_{s,k}(n) := \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}}. \quad (52)$$

Each pole with  $\text{Im } \tau_p \geq 1$  contributes one projected polylog to  $L^*(f, s)$ , whose L-integral we now record.

**Definition 4.11.** For a pole of order  $N+1$  projected at  $z$  with  $\text{Im } z > 1$ , let  $I_N(s, z)$  be the L-integral of the projected polylog  $\lambda_{N,z}$  in the minimal class  $[\hat{\gamma}_f^S], [\hat{\gamma}_f^T]$  of Definition 4.3:

$$I_N(s, z) := L^*(\lambda_{N,z}, s; [\hat{\gamma}_f^S], [\hat{\gamma}_f^T]), \quad (53)$$

where  $\lambda_{N,z}$  denotes the function  $\text{Li}_{-N}(e^{2\pi i(\tau-z)})$ , fed into the linear  $L^*$ -functional.

**Lemma 4.12.** Let  $f \in F_k$  have a pole of order  $p$  at  $z$ ,  $\text{Im } z > 1$ , with projection weights  $r_{f,z}^*(m)$  (Definition 4.5). Polylog inversion  $\text{Li}_{-N}(w) = (-1)^{N+1} \text{Li}_{-N}(1/w) - \delta_{N,0}$  ( $N \in \mathbb{N}_0$ ) resums (53) into (each exponentially convergent, base  $e^{2\pi(1-\text{Im}(z))}$ )

$$L^*(\hat{P}_z f, s; [\hat{\gamma}_f^S], [\hat{\gamma}_f^T]) = \sum_{m=1}^p r_{f,z}^*(m) \frac{(-2\pi i)^m}{(m-1)!} I_{m-1}(s, z), \quad (54)$$

$$I_N(s, z) = \delta_{N,0} \left( \frac{1}{s} + \frac{i^k}{k-s} \right) + (-1)^{N+1} \sum_{n=1}^{\infty} n^N e^{2\pi i n z} G_{s,k}(-n). \quad (55)$$

*Proof.* By Proposition 4.1 we may evaluate  $I_N(s, z)$  in the  $\delta \rightarrow 0$  representative of the minimal class; there the projected polylog, which is regular at  $i$  since  $\text{Im } z > 1$ , has vanishing  $\gamma^S$ -integral and  $\hat{\gamma}_f^T \rightarrow [i-1, i]$ . On  $[i-1, i]$  one has  $|e^{2\pi i(\tau-z)}| > 1$ , so the defining series of  $\text{Li}_{-N}$  diverges; inversion rewrites it as  $(-1)^{N+1} \sum_{n \geq 1} n^N e^{2\pi i n(z-\tau)}$ , convergent since  $|e^{2\pi i(z-\tau)}| < 1$ , and term-by-term integration against  $\tilde{k}_T$  gives  $\sum_n n^N e^{2\pi i n z} G_{s,k}(-n)$ , the mode- $(-n)$  block of Theorem 3.2; for  $N = 0$  the leftover  $-1$  in  $\text{Li}_0(w) = -1 - \text{Li}_0(1/w)$  contributes  $\frac{1}{s} + \frac{i^k}{k-s}$ . The rate is  $e^{-2\pi(\text{Im } z - 1)}$ , from  $\Gamma(s, -2\pi n) \sim (2\pi n)^{s-1} e^{2\pi n}$  against  $|e^{2\pi i n z}| = e^{-2\pi n \text{Im } z}$ .  $\square$

For any integer  $J \geq 1$ , the incomplete gamma satisfies

$$\Gamma(s, X) = e^{-X} \sum_{\ell=0}^{J-1} (s-1)_{(\ell)} X^{s-1-\ell} + (s-1)_{(J)} \Gamma(s-J, X). \quad (56)$$

It peels the leading  $J$  terms of the large- $X$  expansion of  $\Gamma(s, X)$  into elementary powers  $X^{s-1-\ell}$  and collects the rest into an incomplete gamma of lower order  $s-J$  and hence faster-decaying. Importantly, the peeled powers integrate and resum against our kernel in closed form. Choosing  $J$  large enough therefore quarantines the entire singular structure at  $\text{Im}(z) = 1$  into that finite, explicit sum, while the remainder's mode-sum is left manifestly convergent. We use (56) first for the poles on the line  $\text{Im}(z) = 1$  with  $z \neq i$ , and then precisely at  $z = i$ .

Extracting similar results for  $L^*$  evaluated on the polar terms for poles on the line  $\text{Im}(\tau) = 1$  requires more care. To do this away from the special point  $z = i$ , we find it helpful to define the following two functions:

$$\Pi_N(s, z) := \sum_{\ell=0}^N \frac{\text{Li}_{\ell+1-N}(e^{2\pi i(z-i)})}{(-2\pi)^{\ell+1}} \{(s-1)_{(\ell)} + i^k (k-s-1)_{(\ell)}\}, \quad (57)$$

$$R_N(s, z) := \sum_{n=1}^{\infty} n^N e^{2\pi i n z} \left[ (s-1)_{(N+1)} \frac{\Gamma(s-N-1, -2\pi n)}{(-2\pi n)^s} + i^k (k-s-1)_{(N+1)} \frac{\Gamma(k-s-N-1, -2\pi n)}{(-2\pi n)^{k-s}} \right], \quad (58)$$

where  $(a)_{(\ell)} := a(a-1)\cdots(a-\ell+1)$  is the falling factorial ( $(a)_{(0)} := 1$ ),  $\text{Li}_s$  the polylogarithm of Definition 4.5, and  $\Gamma, \Gamma(a, x)$  the gamma and upper incomplete gamma, on principal branches [10].

**Lemma 4.13.** *Let  $f \in F_k$  have a pole of order  $p$  at  $z$  with projection weights  $r_{f,z}^*(m)$  (Definition 4.5). With  $\Pi_N, R_N$  as in (57)–(58), for  $\text{Im } z > 1$  and every  $s \in \mathbb{C}$ ,*

$$L^*(\widehat{P}_z f, s; [\hat{\gamma}_f^S], [\hat{\gamma}_f^T]) = \sum_{m=1}^p r_{f,z}^*(m) \frac{(-2\pi i)^m}{(m-1)!} I_{m-1}(s, z), \quad (59)$$

$$I_N(s, z) = \delta_{N,0} \left( \frac{1}{s} + \frac{i^k}{k-s} \right) + (-1)^{N+1} [\Pi_N(s, z) + R_N(s, z)], \quad (60)$$

and the second line's right side extends  $I_N(s, \cdot)$  continuously to  $\text{Im } z = 1$  for  $z \notin i + \mathbb{Z}$ .

The crux of this Lemma is to show the existence of the smooth continuation of  $I_N(s, z)$  from Lemma 4.12, derived for  $\text{Im}(z) > 1$ , to the  $\text{Im}(z) = 1$ -line away from the elliptic point  $z = i$ . This elliptic point requires special care, which must focus on, below.

*Proof.* We manipulate the main result of Lemma 4.12,

$$I_N(s, z) = \delta_{N,0} \left( \frac{1}{s} + \frac{i^k}{k-s} \right) + (-1)^{N+1} \sum_{n=1}^{\infty} n^N e^{2\pi i n z} \left( \frac{\Gamma(s, -2\pi n)}{(-2\pi n)^s} + i^k \frac{\Gamma(k-s, -2\pi n)}{(-2\pi n)^{k-s}} \right), \quad (61)$$

Our proof proceeds in two steps. First, we isolate the possible sources of divergence on the line  $\text{Im}(z) = 1$  into a finite sum of polylogarithms  $\text{Li}_{\ell+1-N}(e^{2\pi i(z-i)})$  for  $\ell = 0, \dots, N$ . To do this, we apply the recursion (56) to each gamma in  $G_{s,k}(-n)$  at  $X = -2\pi n$ ,  $J = N+1$ . Dividing each incomplete-Gamma term by  $(-2\pi n)^s$  and  $(-2\pi n)^{k-s}$  yields, then, a common factor of  $1/(-2\pi n)^{\ell+1}$  for each term in the finite sum. Further, this finite-sum is proportional to  $e^{2\pi n}$ ; we leave the trailing incomplete gammas as-is:

$$G_{s,k}(-n) = \left\{ e^{2\pi n} \sum_{\ell=0}^N \frac{(s-1)_{(\ell)}}{(-2\pi n)^{\ell+1}} + \frac{(s-1)_{(N+1)} \Gamma(s-N-1, -2\pi n)}{(-2\pi n)^s} \right\} + i^k (s \rightarrow k-s). \quad (62)$$

Now, we consider the  $n$ -sum  $G_{s,k}(-n)$ , starting with the falling factorial terms:

$$\sum_{n=1}^{\infty} n^N e^{2\pi i n z} e^{2\pi n} \left( \sum_{\ell=0}^N \frac{(s-1)_{(\ell)}}{(-2\pi n)^{\ell+1}} + i^k (s \rightarrow k-s) \right) = \sum_{\ell=0}^N \frac{\text{Li}_{\ell+1-N}(e^{2\pi i(z-i)})}{(-2\pi)^{\ell+1}} \{ (s-1)_{(\ell)} + i^k (k-s-1)_{(\ell)} \}, \quad (63)$$

i.e.,  $\Pi_N(s, z)$  of (57), while the reduced-gamma terms remaining are  $R_N(s, z)$  of Eq. (58), which together establish the lemma's identity (60).

The second main step of our proof is to show that  $\Pi_N(s, z)$  is regular as  $\text{Im}(z) \rightarrow 1$  for  $z \neq i$ , while  $R_N(s, z)$  is regular for all  $z$  with  $\text{Im}(z) = 1$ . We treat each piece in turn, beginning with  $R_N$ .

Each term of  $R_N$  is internal to the  $G_{s,k}(-n)$  in the mode  $\sum e^{2\pi i n z} G_{s,k}(-n)$  from Lemma 4.12. Clearly,  $|e^{2\pi i n z}| = e^{-2\pi n \text{Im} z}$ . Thus, the incomplete gamma's exponential growth cancels against the exponentially falling  $e^{2\pi i n z}$  factors. As  $n \rightarrow \infty$ , we have:

$$\left| n^N e^{2\pi i n z} \frac{\Gamma(s-N-1, -2\pi n)}{(-2\pi n)^s} \right| \sim \left| \frac{n^N e^{2\pi i n z} (-2\pi n)^{s-N-2} e^{2\pi n}}{(-2\pi n)^s} \right| \sim \left| \frac{e^{2\pi i n z} e^{2\pi n}}{(-2\pi)^{2+N} n^2} \right| \sim \frac{e^{2\pi n(1-\text{Im}(z))}}{(2\pi)^{N+2} n^2} \quad (64)$$

where the detailed asymptotics of  $\Gamma(s, x)$  in the first scaling relation follow from [10, §8.11]. (The  $i^k(s \rightarrow k-s)$ -partner follows identically.) Hence the  $n$ -th term of  $R_N$  is  $O(n^{-2} e^{-2\pi n(\text{Im} z - 1)})$ , uniformly for  $\text{Im} z \geq 1$ . There  $e^{-2\pi(\text{Im} z - 1)} \leq 1$ , so  $\sum_n n^{-2} e^{-2\pi n(\text{Im} z - 1)} \leq \zeta(2)$ : the series converges absolutely and uniformly, up to and including the edge  $\text{Im} z = 1$ , so  $R_N$  is continuous there.

For  $\Pi_N$ , recall that it is a sum over  $N+1$  polylogarithms of integer index  $\leq 1$ ,  $\Pi_N(s, z) = c_1(s) \text{Li}_1(w) + c_0(s) \text{Li}_0(w) + \dots + c_{1-N}(s) \text{Li}_{1-N}(w)$ , where  $w := e^{2\pi i(z-i)}$ . Each  $\text{Li}_a$  ( $a \leq 1$ ) is single-valued and analytic on the cut plane  $\mathbb{C} \setminus [1, \infty)$ : for  $a \leq 0$  it is rational with a pole of order  $1-a$  at  $w = 1$ , while  $\text{Li}_1(w) = -\log(1-w)$  has a logarithmic branch point there. Every singularity sits at the endpoint  $w = 1$ , and the cut of  $\text{Li}_1$  runs along the ray  $[1, \infty)$ . The edge is the unit circle  $|w| = 1$ , which meets that radial cut only at  $w = 1$ ; approaching it from inside the disc ( $\text{Im} z > 1$ ) gives the analytic boundary value of each  $\text{Li}_a$  at every  $w \neq 1$ . Hence  $\Pi_N$  is analytic along the whole edge except at  $z \in i + \mathbb{Z}$ , where each term diverges. With the continuity of  $R_N$ , (60) seamlessly extends  $I_N(s, \cdot)$  continuously to  $\text{Im} z = 1$  for  $z \notin i + \mathbb{Z}$ .  $\square$

The above result does not apply to singularities at  $z = i$ ; here  $\Pi_N$  is singular ( $w = 1$ ). However, as we will presently show, even though each individual polylogarithm of non-positive index diverges at the point  $z = i$ , the special elliptic nature of that point imposes constraints on the sum of naively divergent terms such that the divergences exactly cancel. However, first, we present an important definition and an associated remark:

**Definition 4.14.** For  $N \geq 0$ , with  $H_N := \sum_{r=1}^N \frac{1}{r}$  and  $c_\ell^+(s) := (s-1)_{(\ell)} + i^k (k-s-1)_{(\ell)}$ , define:

$$\mathcal{E}_N^+(s) := \sum_{n \geq 1} n^N e^{2\pi n} \left[ (s-1)_{(N+1)} \frac{\Gamma(s-N-1, 2\pi n)}{(2\pi n)^s} + i^k (k-s-1)_{(N+1)} \frac{\Gamma(k-s-N-1, 2\pi n)}{(2\pi n)^{k-s}} \right], \quad (65)$$

$$\text{FP}_N(s, k) := \frac{c_N^+(s) H_N}{(2\pi)^{N+1}} + \sum_{\ell=0}^{N-1} \frac{c_\ell^+(s) \zeta(\ell+1-N)}{(2\pi)^{\ell+1}} + \frac{(-1)^N N!}{(2\pi)^{N+1}} \sum_{t=0}^{N-1} \frac{(-1)^t (s-1)_{(t)}}{t! (N-t)} + \mathcal{E}_N^+(s). \quad (66)$$

**Remark 4.15.** The remainders  $R_N$  of Lemma 4.13 (58) and  $\mathcal{E}_N^+$  of Definition 4.14 (65) are what is left after unfolding  $J$  terms of the incomplete-gamma recursion (56); both use  $J = N+1$ . The unfolding isolates the terms that could diverge as  $\text{Im}(z) \rightarrow 1^+$  into an explicit finite set of special functions, and  $J = N+1$  is the least  $J$  that isolates all of them, and leaves a remainder

that converges for every  $z$  on  $\text{Im}(z) = 1$ . Any larger  $J$  isolates the same terms, leaving  $I_N$  and  $\text{FP}_N$  unchanged and merely quickening the remainder's decay ( $\sim n^{-(J-N+1)}$ ) at the cost of carrying further positive-index polylogs  $\text{Li}_2, \dots, \text{Li}_{J-N}$  into the finite sum (contributing values up to  $\zeta(J-N)$  at  $z = i$ ).

**Lemma 4.16.** *Let  $f \in F_k$  have a pole of order  $P$  at  $i$ , with pole weights  $r_{f,i}^*(m)$  (Definition 4.5), i.e. principal part  $\sum_{m=1}^P r_{f,i}^*(m)(\tau - i)^{-m}$ . Modularity  $f(-1/\tau) = \tau^k f(\tau)$  is equivalent to*

$$\sum_{m=r}^P r_{f,i}^*(m) \binom{m}{r} i^m = i^k \sum_{m=r}^P r_{f,i}^*(m) \binom{k}{m-r} i^{-m}, \quad \forall r = 1, \dots, P. \quad (67)$$

The leading relation ( $r = P$ ) is  $((-1)^P - i^k) r_{f,i}^*(P) = 0$ , so the pole order matches  $k \bmod 4$  ( $P$  odd  $\Leftrightarrow k \equiv 2$ ,  $P$  even  $\Leftrightarrow 4 \mid k$ ); at  $P = 1$  this is  $(1 + i^k) r_{f,i}^*(1) = 0$ , and for a double pole the  $r = 1$  relation is  $r_{f,i}^*(1) = \frac{i(k-2)}{2} r_{f,i}^*(2)$ .

*Proof.* Write  $\tau = i + \varepsilon$ . Then  $-1/\tau - i = -\varepsilon/(1 - i\varepsilon)$ , so the principal part of  $f(-1/\tau)$  is  $\sum_m r_{f,i}^*(m)(i - \varepsilon^{-1})^m$  and that of  $\tau^k f(\tau)$  is  $i^k (1 - i\varepsilon)^k \sum_m r_{f,i}^*(m) \varepsilon^{-m}$ . Matching the coefficient of  $\varepsilon^{-r}$  through  $(i - \varepsilon^{-1})^m = \sum_j \binom{m}{j} i^{m-j} (-1)^j \varepsilon^{-j}$  and  $(1 - i\varepsilon)^k = \sum_l \binom{k}{l} (-i)^l \varepsilon^l$  gives  $(-1)^r \sum_{m=r}^P r_{f,i}^*(m) \binom{m}{r} i^{m-r} = i^k \sum_{m=r}^P r_{f,i}^*(m) \binom{k}{m-r} (-i)^{m-r}$ ; dividing through by  $i^r$  (with  $-1 = i^2$  and  $-i = i^{-1}$ ) yields (67).  $\square$

**Lemma 4.17.** *Let  $f \in F_k$  have a pole of order  $P$  at  $i$  with pole weights  $r_{f,i}^*(m)$  (Definition 4.5), and  $\hat{P}_i f$  its projection. Evaluated in the minimal class at basepoint  $i(1 + \delta)$ , the pole's contribution to  $L^*(f, s)$  has the finite  $\delta \rightarrow 0$  limit*

$$L^*(\hat{P}_i f, s) := \lim_{\delta \rightarrow 0} L^*(\hat{P}_i f, s; i(1 + \delta)) = \sum_{m=1}^P \frac{(-2\pi i)^m r_{f,i}^*(m)}{(m-1)!} \text{FP}_{m-1}(s, k), \quad (68)$$

entire in  $s$ . The limit exists precisely because the pole weights obey the relations of Lemma 4.16 at the  $S$ -fixed point  $i$ : they cancel the individually divergent contour pieces.

*Proof.* We begin by noting that, at its branch point  $w = e^{2\pi i(\tau-i)} \sim 1$  the polylogarithm expands as  $\text{Li}_{1-m}(e^{2\pi i(\tau-i)}) = (m-1)! (-2\pi i(\tau-i))^{-m} + \sum_{p \geq 0} \frac{\zeta(1-m-p)}{p!} (2\pi i(\tau-i))^p$  ([10, §25.12]). Thus,

$$\frac{(-2\pi i)^m}{(m-1)!} \text{Li}_{1-m}(e^{2\pi i(\tau-i)}) = (\tau-i)^{-m} + \frac{(-2\pi i)^m}{(m-1)!} \sum_{p=0}^{\infty} \frac{\zeta(1-m-p)}{p!} (2\pi i(\tau-i))^p, \quad (69)$$

is a pure order- $m$  pole, as in Lemma 4.6, plus an entire,  $T$ -periodic tail. Hence  $\hat{P}_i f = \sum_{m=1}^P r_{f,i}^*(m)(\tau - i)^{-m} + (\text{regular at } i)$ . As the contour  $\text{Im}(\hat{\gamma}_f^T)$  has height  $1 + \delta > 1$ , the analogous step of  $w$ -inversion,  $\text{Li}_m(w) = -\delta_{m,0} - (-1)^m \text{Li}_m(1/w)$  from the proof of Lemma 4.12 does not need to be invoked. Thus there is no  $\frac{1}{s} + \frac{i^k}{k-s}$  in the final result.

We now extract their finite parts of this polylogarithm on the  $\hat{\gamma}_f^T$  and  $\hat{\gamma}_f^S$  contour-integrals.

On  $\hat{\gamma}_f^T$  the block is the convergent series  $\text{Li}_{1-m}(e^u) = \sum_{n \geq 1} n^{m-1} e^{2\pi n} e^{2\pi i n \tau}$ ; integrating against  $\tilde{k}_T$  from the basepoint  $\tau_0 = i(1 + \delta)$  gives, by Theorem 3.2,  $\sum_{n \geq 1} n^{m-1} e^{2\pi n} \left[ \frac{\Gamma(s, X_n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, X_n)}{(2\pi n)^{k-s}} \right]$  with  $X_n := -2\pi i n \tau_0 = 2\pi n(1 + \delta)$ . Expanding by the recursion (56) at  $J = m$ , and using

$e^{2\pi n}e^{-X_n} = e^{-2\pi n\delta}$  and  $X_n^{s-1-\ell} = (2\pi n)^{s-1-\ell}(1+\delta)^{s-1-\ell}$ , the  $\ell$ -sum becomes (with  $c_\ell^+(s)$  collecting the  $s$ - and  $i^k(k-s)$ -parts)

$$\sum_{\ell=0}^{m-1} \frac{c_\ell^+(s)(1+\delta)^{s-1-\ell}}{(2\pi)^{\ell+1}} \text{Li}_{\ell+2-m}(e^{-2\pi\delta}) + \mathcal{E}_{m-1}^+(s), \quad (70)$$

$\mathcal{E}_{m-1}^+$  being the reduced- $\Gamma$  remainder of (65). With  $\text{Li}_{\ell+2-m}(e^{-2\pi\delta}) = \Gamma(m-\ell-1)(2\pi\delta)^{\ell-m+1} + \sum_{p \geq 0} \frac{\zeta(\ell+2-m-p)}{p!}(-2\pi\delta)^p$ , two finite ( $\delta^0$ ) parts survive:

1. The regular part at  $\delta^0$  gives, for  $\ell \neq m-1$ , the  $\zeta$ -value  $\frac{c_\ell^+(s)}{(2\pi)^{\ell+1}} \zeta(\ell+2-m)$ : the  $\zeta$ -sum of (66).
2. For  $\ell < m-1$  the singular  $\Gamma(m-\ell-1)(2\pi\delta)^{\ell-m+1}$  crosses the  $\delta^{m-\ell-1}$  coefficient  $\binom{s-1-\ell}{m-\ell-1}$  of  $(1+\delta)^{s-1-\ell}$ ; by the falling-factorial identity  $(s-1)_{(\ell)}(m-\ell-2)!\binom{s-1-\ell}{m-\ell-1} = (s-1)_{(m-1)}/(m-\ell-1)$  (and its  $k-s$  partner) these sum to

$$\frac{c_{m-1}^+(s)}{(2\pi)^m} \sum_{\ell=0}^{m-2} \frac{1}{m-\ell-1} = \frac{c_{m-1}^+(s) H_{m-1}}{(2\pi)^m}, \quad (71)$$

the harmonic term of (66).

The remaining  $\ell = m-1$  term is just  $\log(2\pi\delta)$ , which cancels in the  $\delta \rightarrow 0$  limit of the S-contour. We now turn to this S-contour's contribution.

On  $\hat{\gamma}_f^S$  the pure pole  $\sum_m r_{f,i}^*(m)(\tau-i)^{-m}$ , integrated by principal value against  $\tau^{s-1}$  along the axis ( $\tau = i(1+v)$ ), has the  $\delta^0$  finite part

$$i^{-m} \text{PV} \int \frac{(1+v)^{s-1}}{v^m} dv \Big|_{\delta^0} = i^{-m} (-1)^{m-1} \sum_{t=0}^{m-2} \frac{(-1)^t (s-1)_{(t)}}{t! (m-t-1)}, \quad (72)$$

the polynomial term of (66). Collecting, the finite part of block  $m$  is  $\frac{(-2\pi i)^m}{(m-1)!} r_{f,i}^*(m) \text{FP}_{m-1}(s, k)$ .

*Divergences cancel by (67).* The residual pieces are the  $\delta \rightarrow 0$  divergences  $\delta^{-(m-1)}, \dots, \log \delta$  from the pole part  $\sum_m r_{f,i}^*(m)(\tau-i)^{-m}$  at the  $S$ -segment endpoints straddling  $i$ . At order  $r$  their coefficient is

$$\sum_{m=r}^P r_{f,i}^*(m) \binom{m}{r} i^m - i^k \sum_{m=r}^P r_{f,i}^*(m) \binom{k}{m-r} i^{-m}, \quad (73)$$

precisely the difference of the two sides of the modularity relation (67), hence 0 for  $f \in F_k$ . Every  $\delta^{-r}$  and  $\log \delta$  term therefore cancels and the  $\delta \rightarrow 0$  limit exists. Conversely, weights violating eq:Laurent at order  $r$  leave that piece uncanceled (at  $r = 1$ , a  $\log \delta$ ), so the limit exists only for  $f \in F_k$ .

The finite part and the (vanishing) divergences give (68).  $\square$

**Theorem 4.18.** *Let  $f \in F_k$  have poles  $\tau_p$  of order  $P_p$  with  $\text{Im } \tau_p \geq 1$  in the strip  $|\text{Re } \tau| \leq \frac{1}{2}$ , pole-projected Fourier coefficients  $c_{Rf}(n)$  (Definition 4.7), and projection weights  $r_{f,\tau_p}^*(m)$  (Definition 4.5). In the minimal class,*

$$L^*(f, s) = -c_{Rf}(0) \left( \frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_{Rf}(n) G_{s,k}(n) + \sum_{\tau_p \neq i} \sum_{m=1}^{P_p} r_{f,\tau_p}^*(m) \frac{(-2\pi i)^m}{(m-1)!} I_{m-1}(s, \tau_p) + L^*(\hat{P}_i f, s), \quad (74)$$

with  $I_{m-1}$  from Lemmas 4.12–4.13 and the last term (present only for a pole at  $i$ ) from Lemma 4.17. The right side is entire in  $s$  apart from at most simple poles at  $s = 0$  and  $s = k$ , and reduces to the weakly-holomorphic  $L^*$  of Theorem 3.2 when  $f$  is holomorphic on  $\mathbb{H}$ .

*Proof.* Split  $f = (I - \hat{P})f + \hat{P}f$ . By Lemma 4.10 the Fourier series of the pole-projected part converges on  $\hat{\gamma}_f^T$ ; Theorem 3.2 applied term by term gives its  $L^*$  as the first two sums. The projection  $\hat{P}f = \sum_{\tau_p} \sum_m r_{f,\tau_p}^*(m) \frac{(-2\pi i)^m}{(m-1)!} \text{Li}_{1-m}(e^{2\pi i(\tau-\tau_p)})$  contributes, for each  $\tau_p \neq i$ , the  $L$ -integrals  $I_{m-1}(s, \tau_p)$  of Definition 4.11, evaluated by Lemmas 4.12–4.13; the pole at  $i$  contributes  $L^*(\hat{P}_i f, s)$  of Lemma 4.17. Entirety and the pole locations follow from those of  $G_{s,k}$  and the  $\delta_{N,0}$  terms; when  $f$  is holomorphic on  $\mathbb{H}$ ,  $\hat{P}f = 0$  and  $c_{Rf} = c_f$ , recovering Theorem 3.2.  $\square$

**Remark 4.19.** Casual inspection of the structure of the  $I_N(s, \tau_p)$ -functions entering into Theorem 4.18 would lead one to conclude that they substantively differ from those in [3], where the author first defined an  $L$ -function for  $f \in F_k$ : Ref. [3] has the  $1/s$ -residue proportional only to  $c_f(0)$ . However, our  $L$ -function has  $1/s$ -residue contributions from all simple poles located above the  $\text{Im}(\tau_p) > 1$ -line. This naive difference is immaterial, and does not impact the results of the residue theorems for the  $1/s$ -poles in [3], for the simple reason that in Ref. [3], the reference contour was implicitly at  $\tau_0 \rightarrow i\infty$ . This was valid in the restricted context of [3], as that paper only focused on  $f \in F_k$  where  $f = \sum_{n \geq 0} c_f(n)q^n$ ; poles at the  $\text{SL}_2(\mathbb{Z})$  cusp were explicitly excluded, which allowed the limit  $\tau_0 \rightarrow i\infty$  to be smooth and safe. However, in this work, we make a different reference contour — that in Definition 4.3.

**Remark 4.20.** The  $1/s$  residue is carried entirely by the  $T$ -segment:  $L_S^*$  is a finite integral of  $f\tau^{s-1}$  and is entire in  $s$ , while  $\tilde{k}_T$  has a simple pole at  $s = 0$  whose coefficient is  $\int_{\hat{\gamma}_f^T} f d\tau$ , the constant Fourier mode of  $f$  in the strip containing that segment. Deforming  $\hat{\gamma}_f^T$  to pass above every pole of  $f$  therefore returns  $\text{Res}_{s=0} L^*(f, s) = -c_f(0)$ , the constant term at the cusp, and recovers the residue bookkeeping of [3]. That class differs from the one of Definition 4.3 by the crossings of Lemma 4.4.

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